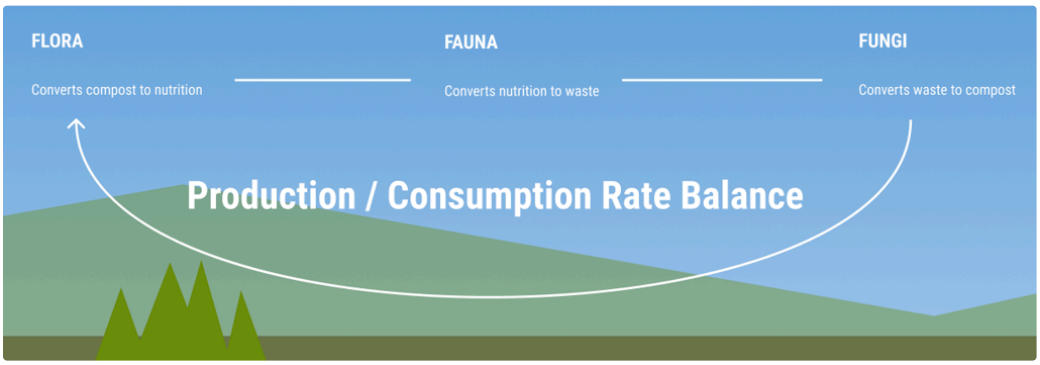
⚖️ Harmony System

Purpose

The player must manage and maintain the balance between flora, fauna, and fungi to spawn new dinosaurs.

Mechanics

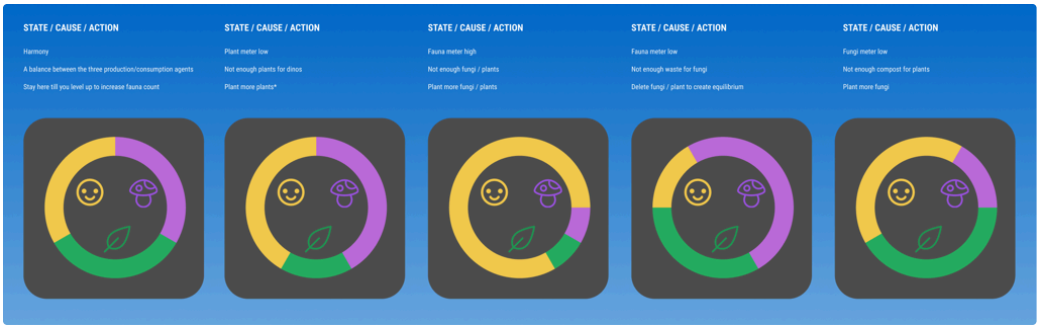


Each actor (flora, fungi, fauna) contributes points/resources to the ecosystem that balance +/- each other:

	🦖Baby Triceratops (1)	🌿Fern (3)	🍄Basic Fungi (2)
Consumption	100 NPM	100 CPM	100 WPM
Production	100 WPM	100 NPM	100 CPM

- NPM = **Potential** Nutrients Per Minute = (Nutrient Production/Tick)*60
  - Produced by flora, consumed by fauna.
- CPM = **Potential** Compost Per Minute = (Compost Production/Tick)*60
  - Produced by fungi, consumed by flora.
- WPM = **Potential** Waste Per Minute = (Waste Production/Tick)*60
  - Produced by fauna, consumed by fungi.

We use the “Potential” number so that the HUD becomes *responsive* to the player’s actions and communicates to the player whether they need to plant more plants or fungi or not (i.e. visualize where they would need to put their efforts with a quick glance):



The player can react to the size of slices and better understand where the imbalance is and how far they are from reaching "Harmony".

Harmony

"Harmony", notice the slices are equal.

When the HUD indicates balance, the player progress towards levelling up. When the player levels up, a new dinosaur will enter the habitat, which throws off the balance and gives the player a new target to work towards.

Link to **HUD / Harmony Meter** specifications:

- HUD

Parameters

Play with the pie chart here:

DH - Points and Progressions

BETA Iterations

When in harmony bonus

Purpose

Currently, when in harmony the player is effeciently earning XP but there is nothing for the player to do while this happens other than to sit and wait for the bar to fill up.

Mechanics

To make this special, when in full harmony:

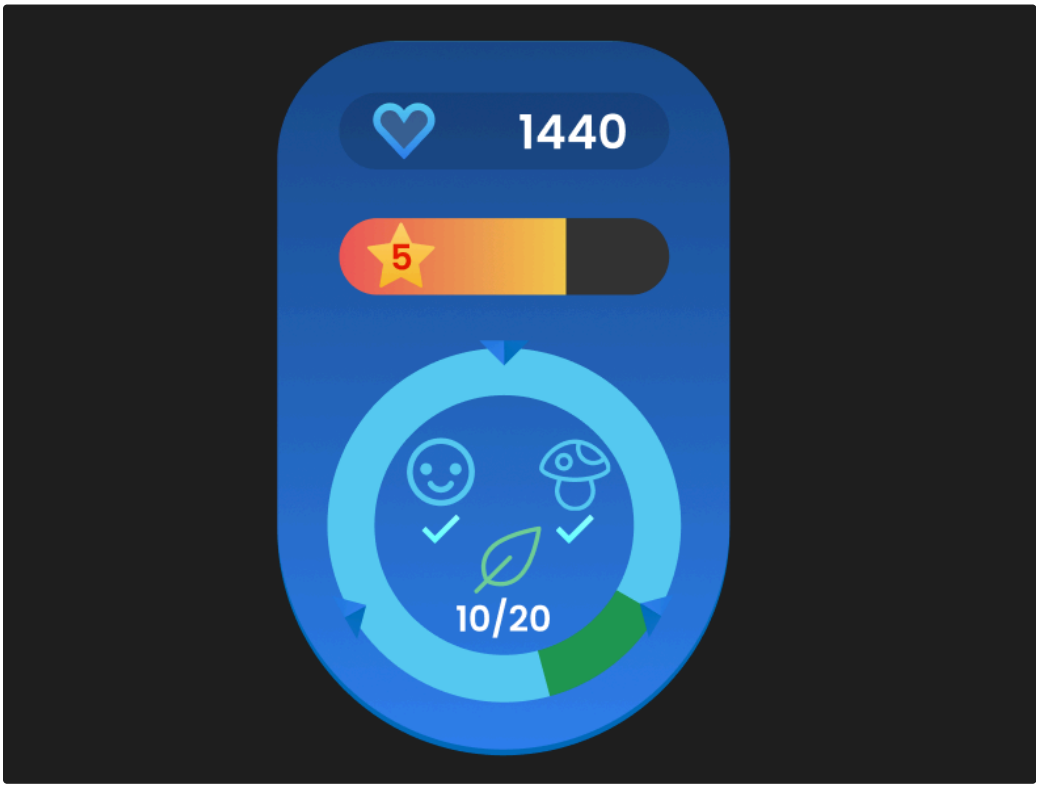
- the dino DP is increasing by a multiplier (e.g. 10 times as fast) so the player is now busy collecting DP before that state ends (when a new dinosaur appears).
- the dino dances after peing petted.
- ~~the player get double the DP when petting a dino.~~
- LUT to increase vibrancy as a post-FX.

Parameters

BP_EnvironmentHarmonyHandler (Blueprint Class)

- Dino DP generation **mutliplier** when in Harmony. Adjustastable by designer in the BP_EnvironmentHarmonyHandler class.

Harmony wheel



One weird but somewhat expected problem is that the player can play too well and plant too much, thus being unable to progress as they are in "disharmony." I thought there was a way to balance the game to prevent this, but with all the variables that are a play, I realized that was wishful thinking. There are a few solutions to this (priority high to low):

- This is an old idea since the beginning of the project; have it so that each player level has different requirements. Getting to level 2 requires at least 40 production points for each (the

pie chart would look balanced at the minimum requirement). Once the minimum requirement is met, the respective piechart slice gets blue, and the respective icon gets a checkmark. This allows the player to plant too much yet not be punished for playing too fast.

- For example, at level 1, let's say the player needs a balance of 40/40/40 and they have
  - 1 Dino with 40 waste points (blue and check mark)
  - Several plants with 80 plant points (blue and check mark)
  - 1 fungi with 20 compost points (still purple, no check mark)
  - This would mean they are 2/3 in Harmony, and the level up meter would still go up, but not as fast as it would if the player was 3/3 in Harmony (as described in the bullet point below). We still want the player to be incentivized to balance 3/3 in Harmony, so the speed at which they level up and get their next Dino is a variable configurable by design, so 2/3 in Harmony could still be rather slow compared to 3/3 in Harmony. **This will be for design to balance and figure out what speeds/time works best.**
- Willis suggests that the player always progresses toward the next level, and the pace of XP accumulation increases when in harmony. There needs to be more to solve the main problem: the game punishes the player for investing in too nutritious food for the dinosaurs/habitat. But it ensures the game keeps adding dinos, so the player must be prepared for the incoming arrivals which keeps the game progressing
- Add an adjustable tolerance threshold in the harmony handler blueprint so the designer can adjust how close to 0.3333, 0.3333, and 0.3333 is considered "harmony".

Note: full harmony appears to be achieved and a level is complete when the player plants about 30 plants per level, afterwards if the player plants more the performance drops

For T-Rex Level (Alpha Test Solution):

- Triceratops only spawn according to flora and fungi harmony (if they're above XX%)
- T-rex only spawn according to fauna harmony (if it's above XX% - and if possible only takes into account T-Rex satisfaction)

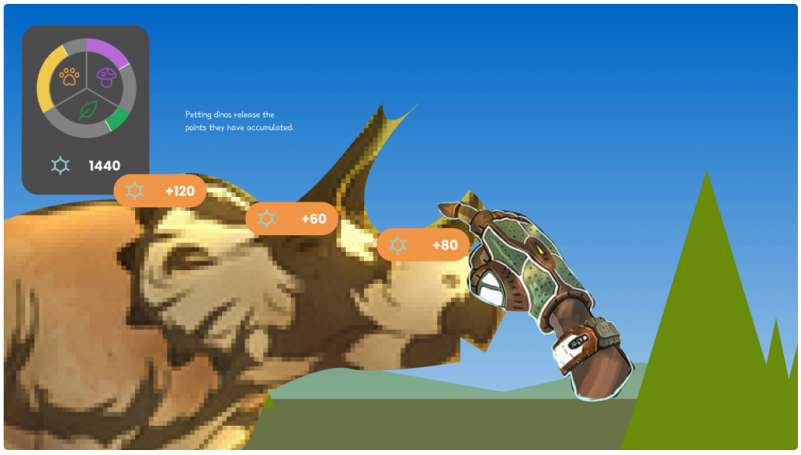
❌ *DEPRECATED BELOW*

Finding Harmony

Essentially, the player has these three components they have to manage:

- **Flora - Nutrients (are there enough total nutrients stored in plants in the environment relative to the total Dino Hunger demand)**
- **Fauna - Dino Average Satisfaction (measure of average happiness among all dinos - determined by when each individual dino checks/meets its needs)**
- **Fungi - Compost (is there enough compost total in environment relative to total plant nutrient capacity) (post VS)**

Flora, Fauna, Fungi Wheel of Harmony



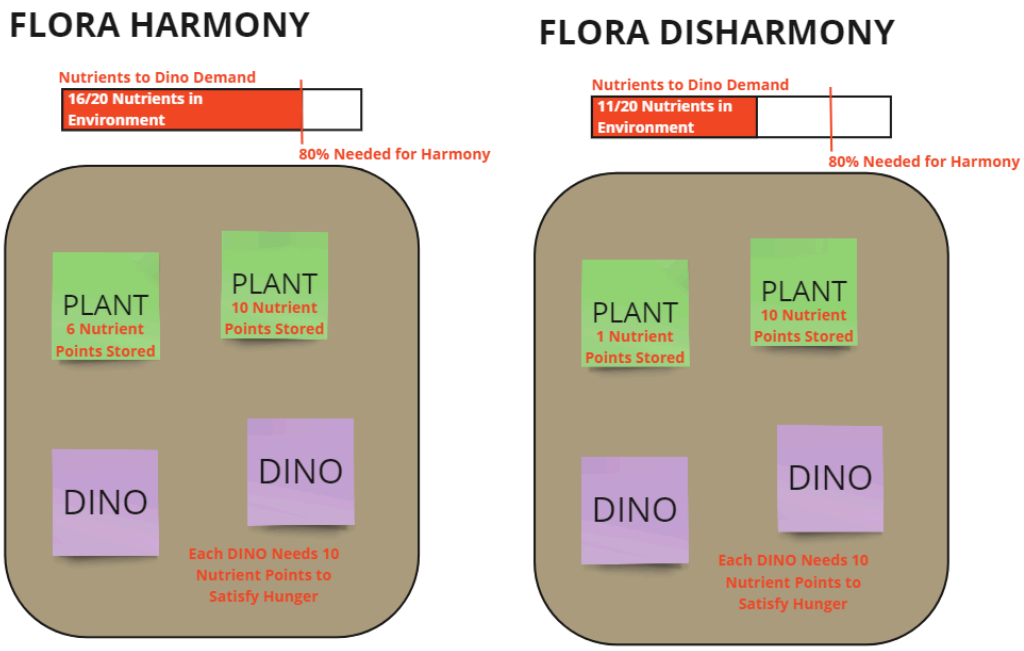
Flora, Fauna, Fungi Wheel of Harmony via HUD

Flora/nutrients

For there to be balance in the environment flora, first, there needs to be “enough” Nutrient points produced by flora to provide for the fauna (specifically dinos).

For example, there are two dinos are in the environment and each require 10 nutrient points (hunger) each to be satisfied and the threshold for Flora s 80%: therefore the environment needs at least 16 nutrient points stored in the environment (across any number of plants).

A nutrient points bar is the detailed view of this section, where the total max of the bar is the total dino demand and how many points stored is the value of nutrients across all plants stored in the environment. If the player fills this bar the **flora (green) section of the Harmony Meter lights up.**

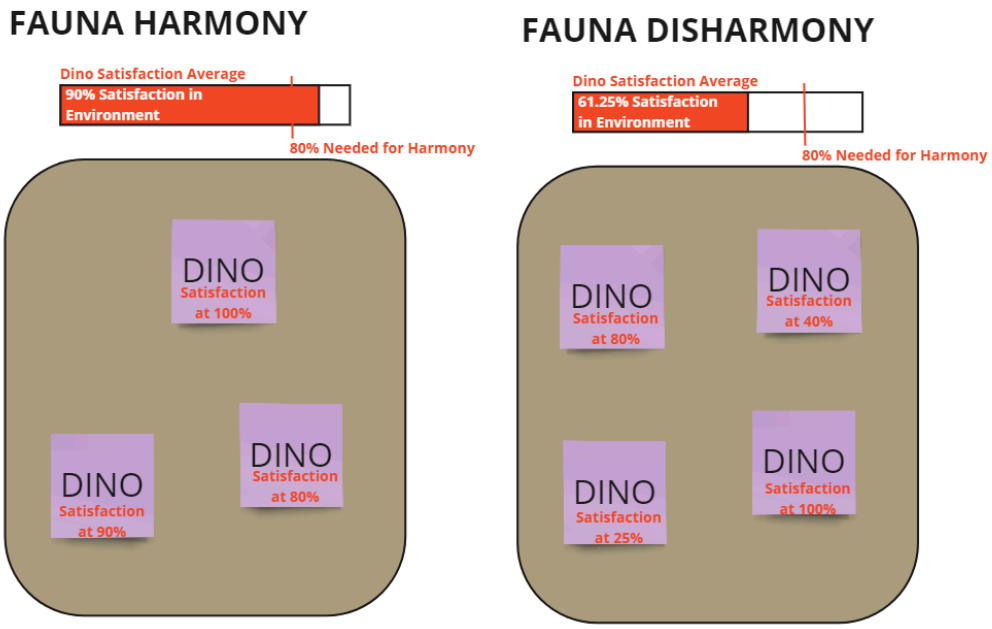


Flora Harmony and Disharmony

Fauna

The total Dino average happiness needs to reach a certain threshold for the **fauna (yellow) section of the Harmony Meter to light up.**

For example, if two dinos are in the environment: one dino is satisfied after all its individual nutrient and cover checks, its happiness is 100% (5/5). The second dino isn't able to satisfy both checks and is reduced to 50% happiness. The average happiness is therefore 75%. If the threshold for the Dino Average Happiness bar is 80%, then the yellow section of the harmony meter will not light up.



Fauna Harmony and Disharmony

Fungi (post VS)

Fungi needs to be balanced. Consumers naturally produce dung. Dinos will produce dung at a rate of X per Y time period. The dung will be dropped by dinos around the map, but with a preference to be dropped near Fungi networks.

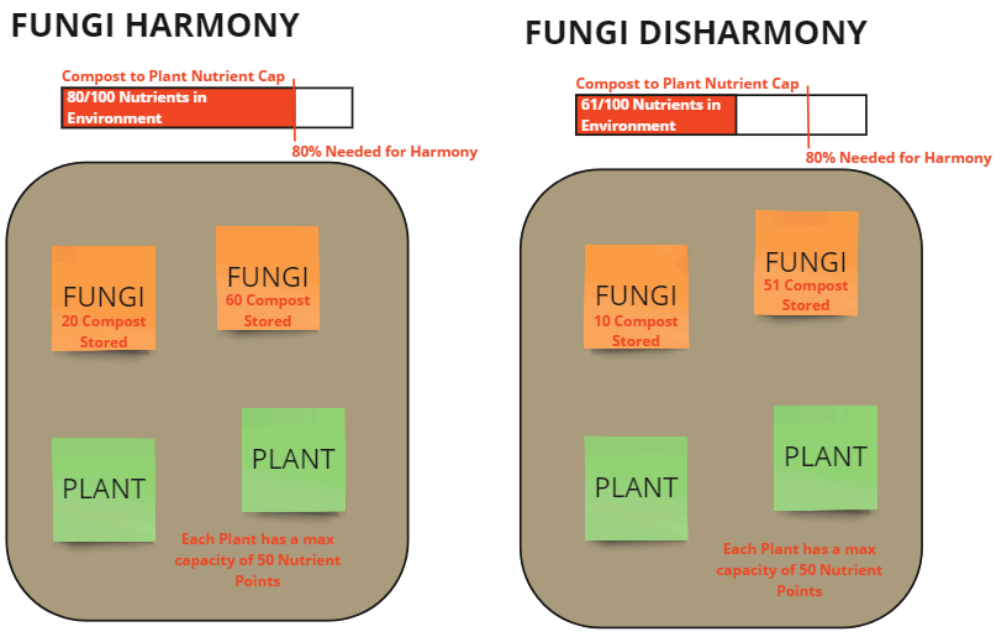
Similar to how Plants store Nutrients within themselves, the **Dung will have a set amount of Waste Points stored within themselves**.

Unlike plant Nutrient Points, Waste Points are by themselves not useful. Therefore the player must take measures to convert Waste Points into useful Compost Points using Fungi networks.

Dung that is nearby Fungi will be slowly decomposed into stored Nitrogen, which Plants then use to convert into Nutrient points at a 1:1 ratio. The player can use the Manipulator Tool to move the dung to Fungi networks (both the dung and fungi asset will have a radius and if the radius is placed overlapping each other, then it would be considered “nearby”).

If the amount of Compost Points in the environment meets a **XX% threshold** of the **total nutrient capacity of plants** the **fungi (purple) section of the Harmony Meter lights up.**

Once, all three pieces of the Harmony Meter are lit up/at Harmony threshold, is the ecosystem considered in Harmony. And of course, the different parts of the ecosystem can move in and out of Harmony (i.e., the pie chart can light up and un-light).



Fungi Harmony and Disharmony

Consequences for Disharmony

- The consequences for Disharmony are naturally built into the mechanics of each ecosystem actor (Flora, Fauna and Fungi), if any part of the ecosystem isn't lit up/achieved Harmony, eventually Dinos will become unsatisfied, sick and leave.
- New dinosaurs are also less likely to appear when the environment is in Disharmony. More dinosaurs are attracted to a harmonious environment. The amount of dinosaurs determines the player level.

Harmony Metrics	DESCRIPTION	Interacts with	Example Value
Nutrient Points in Environment	The total number of nutrient points stored among flora in the environment.	Plants Nutrient Points Stored	20 total nutrient points (e.g., 1 plant = 10 nutrient points + 1 fungi = 10 nutrient points)
Nutrient Bar Capacity	Total max capacity of Nutrient Demand in the environment. Determined by the number of dinos	Dino Hunger	40 total (e.g., 2 dinos x 20 required nutrients points)

	and their **Hunger** summed up.		
Nutrient Bar Harmony Threshold	The minimum percentage requirement of **Total Nutrient Points in Environment** out of **Nutrient Bar Capacity** that's needed to for Flora to be in Harmony.	--	>80%
Dino Satisfaction Total Average	The total average Satisfaction of Dinos in the environment.	Dino Satisfaction	70%
Dino Satisfaction Max	Total average capacity for satisfaction of dinos. Essentially equal to satisfaction of an individual Dino.	Dino Satisfaction	100%
Dino Average Satisfaction Harmony Threshold	The minimum percentage requirement of **Dino Satisfaction Total Average** out of **Dino Satisfaction Average Capacity** that's needed to for Fauna to be in Harmony.	--	>80%
Fungi Compost in Environment (post VS)	The total number of Compost Points stored among dung in the environment.	Fungi	20 total compost points (e.g., 1 fungi = 7 compost + 1 fungi = 13 compost)
Fungi Bar Capacity (post VS)	Total capacity of Nutrients in the environment. Determined	Plants Total Nutrient Capacity	100 total (e.g., 2 plants x 50 max nutrient value)

	by the number of plants and their **Max Nutrient Value** summed up.		
Fungi Bar Harmony Threshold (post VS)	The maximum percentage of **Total Fungi Compost in Environment** out of **Fungi Bar Capacity** that's needed to for Fungi to be in Harmony.	--	>80%

Actions

- [ ] Flora/Nutrients harmony is reached when there enough total nutrients stored in plants in the environment relative to the dino total demand
- [ ] Fauna/Satisfaction harmony is reached when average happiness among all dinos reaches an average satisfaction threshold.
- [ ] Fungi/Compost harmony is reached when there is enough compost relative to total plant nutrient capacity in the environment.
- [ ] Overall environment harmony is reached when the three Flora, Fauna and Fungi harmony have been reached.

Stats

The main stats of concern to the player are:

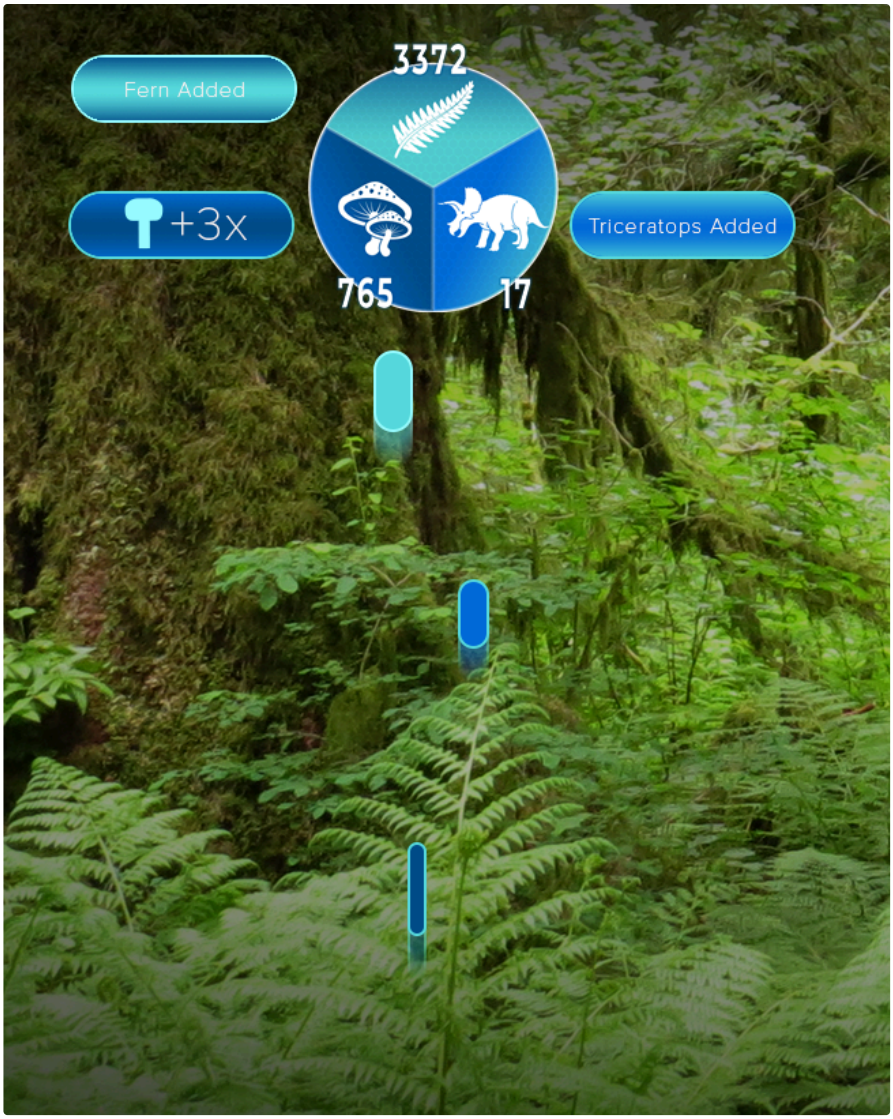
- **Each Flora produce one or more of the following:**
  - X amount of Nutrient Points over time (see Flora for more details)
  - X amount of Cover Points (see Flora for more details)
- **Each Fauna (specifically Dinos)**:
  - Dinos produce X amount of Dino Points over time (used as currency, but does not directly play into environmental Harmony)
  - Dinos consumes by performing "checks" in the environment
    - Nutrient Point Check
      - Dino goes to plant(s) and consumes the nutrient points the plants have generated until it reaches its hunger threshold. It will keep going to other plants until it's full. If it can't consume above the threshold, the check fails.
    - Cover Check
      - Dinos have a cover value that is needed (Required Cover per Check) if there is at least one type of plant that satisfies its cover value, the dino will rest there and sleep, if it can't find one the check fails.
    - If Dino's checks succeed or fail

        - That individual dino's satisfaction goes up or down respectively. Satisfaction of a Dino determines whether it will stay or leave the environment.
    - Dinos performs checks every when their internal Hunger (for Nutrients) and Rest (for Cover) values drop below a specific threshold
  - Produce X amount of Dung time - which produce X amount of Compost Points over time **(post VS)**
- **Each Fungi: (post VS)**
  - Convert Waste Points from nearby (within X radius) Dung into Compost Points (which plants will use to create Nutrient Points at a ratio of 1:1)

Art

The following is a proposal for visually communicating:

- That **the Ecosystem is dynamically updating ALL THE TIME**
- Environment variables visually contributing to the Harmony Meter in real time via **animating 'bubbles'**
- **'Pop Out' information cells** for each of the 3 sections convey specifically WHAT changed with icons, text and numbers
- **Real time numeric totals** displayed for each of the 3 sections - Flora, Fauna, Fungi
- **Real time proportional scaling** for each of the 3 'Pie' sections / wedges
- **'Pie' animated rebalancing at prescribed intervals** with proportionally largest wedge weighted to the bottom



Flora, Fauna, Fungi Harmony Meter

Detailed Breakdown

- **Environment variables** visually contribute to harmony Meter updates
  - These can be **Player-driven or EcoSystem-driven** as the environment naturally ebbs and flows due to a variety of factors
  - They are **classified by the 3 key variables for EcoSystem balance / Harmony** - Plants, Dinos, Fungi
  - As conditions continuously change in the environment **'bubbles' can be seen moving / animating 'from around the environment' in smooth arcing paths up to the Harmony Meter** in the upper left corner of the HUD
  - These elements are **visually discreet and subtle**. If this system is continually updating these elements should visually reinforce the dynamic nature of the system but avoid drawing too much attention

    - These **paths and origins are largely random and abstract representations** - indicative of the dynamically recalibrating EcoSystem at large - but **any real time Player edits should have these elements emit from the specific contextual location** that the event occurs
    - **This feature may be too cumbersome for Oculus performance** and it may be prudent to rely on the following features instead
- **'Pop Out' information cells** for each of the 3 sections to confirm that SOMETHING CHANGED and what specifically changed via icons, text and numbers
    - These **cells are associated with their respective sections of the Harmony Meter** via onscreen alignment / position / orientation / proximity, colour and iconography
    - **The cells 'pop out' / animate** from behind the Harmony Meter as the value gets added, remain onscreen briefly and then quickly disappear / fade out
    - It is possible to have **multiple cells updating simultaneously or in rapid succession**. In this case the **Player's stats should ALWAYS have priority over any others**.
    - How much detail is required for these? Do we need to see / identify specific Plant, Dino or Fungi species? I think this would be ideal as it draws a strong connection between the player's actions and the results they have.
    - Ideally all potential information here to be conveyed in all categories is **clearly classified into concise and meaningful 'phrases'** or blurbs
        - i.e. **'Triceratops added x2'** (this suggests 2 dinos have entered the Level due to the Player's efforts to make the area attractive to them
        - i.e. **'Date Palm added x6'** (this could suggest plant self-propagation or Player actions that occur over short window)
- **Real time numeric totals** displayed for each of the 3 sections - Plants, Dinos and Fungi
    - These **cells are associated with their respective sections of the Harmony Meter** via alignment / position / orientation / proximity
    - No idea how we calculate these totals or how valuable they may be to the Player other than to show more explicit data / values. The goal here was also to show that the Harmony Meter is a PROPORTIONAL indicator ONLY. The size of the 'pie wedges' is proportional to the totals regardless of what those totals are. The size of the 'pie' / harmony Meter never changes.

Optional Aggregate Indicators

- Aggregate Indicators **convey the realtime ebb and flow** within the environment between Harmony Meter update intervals
- 'Hashmarks / buckeyes/ icons are **added or removed relative to realtime EcoSystem updates**
  - i.e. Player plants 20 ferns
  - i.e. Dinos eat 20 ferns
  - i.e. 3 Dinos leave the area
  - i.e. 75 mushrooms propagate on DinoDung
- Each hashmark **can indicate an aggregate value** for each component type
  - i.e. 1 hashmark = 1 Dino
  - i.e. 1 hashmark = 25 Fungi
  - i.e. 1 hashmark = 5 Plants
  - etc.
- These could potentially replace the explicit numerical values described above which might be visually more pleasing / less cluttered
- As the window interval terminates the hashmarks can animate to stretch upwards from their respective bases around the perimeter of the meter than sequentially slide / collapse to the left / clockwise as they feed into the ‘pie slice’ and the whole Meter rebalances. Could be a really nice moment with a strong visual cue and animation as well as a distinctive but subtle audio signature. i.e. chime, stinger, etc.

Background Environment Art Reflecting Harmony and Disharmony

In Virtual Reality, background plants are plants that are not interactable (non-hero plants) with the player/player actions. These plants are along the border of the playable map area, where the player cannot access via locomotion, but can still see.

The background environment colors, level of dynamism, density of background plants, and animal activity should still visually reflect the state of harmony/disharmony.

If the environment is in harmony (all three pieces of the pie are checked/lit up) the background environment should appear more vibrant, colorful, and dynamic. Greater saturation in background colours, trees and leaves rustling, greater leave density, and other animal life can appear.

If the environment is in disharmony, the opposite should happen. The saturation of the background should be lower, trees and leaves move less, lesser leave density, and less sign of animal life.

Background foliage should become denser as players level up. As the player levels up (because they are able to continually achieve harmony), the background environment should become more full with foliage. A Level 1 player will have few background plants and trees, while Level 18 players will have a denser ecosystem.

1. We can potentially achieve all this via:
    a. Global color channel parameters
    b. Plant propagation
    c. Animal entry / dispersal to the area is probably best accomplished via static meshes of mostly adult animals that translate / slide into or out of the play areas. Adding animation could be a stretch goal
    d. Background plant leave density

Prototype (Vertical Slice) Art

Just for Vertical Slice, since we will only have Dev Art UI for the Harmony Meter, we want to update it to be a little more clear to the player what they need to do. This will require some cosmetic changes to the Dev UI Harmony Meter (the calculations are the same).

First, please update the threshold for harmony to be 50% (no longer 80%). Therefore, when they are above 50%, they are considered in Harmony.

There is a **solid threshold line that indicates that 50% is the target** and it sits in the middle of each bar. The bar filling will be attached to this threshold line. For example, the Plant Nutrition in the image above indicates that Plant Nutrients relative to Dino Demand is at about 80% full. Average Dino Satisfaction in the Image above indicates that it is at 30% (about 20% less than 50%).

Post VS Alternative Proposal for Harmony (if current system doesn't test well)

The original idea behind the pie chart was for the player to have visualized where they would need to put their efforts with a quick glance, so the math would be something like this:

- Plant production (nutrients) - Dino consumption (nutrients) = FLORA POINTS (FL)
- Dino production (waste) - Fungi consumption (waste) = FAUNA POINTS (FA)
- Fungi production (compost) - Plant consumption (compost) = FUNGI POINTS (FU)

FL+FA+FU=XFL/X=floraSlice
FA/X=faunaSlice
FU/X=fungiSlice

The player could then react to the size of slices to better understand where the imbalance is and how far they are from reaching harmony.

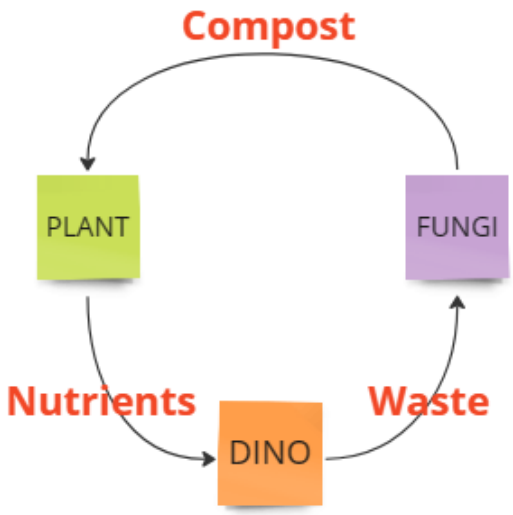
Class Properties:

`CompostAmount: float` //Increased by Dinosaur's Dung

`ListOfFloraActors: [Plants]` //To know when events happen.

`ListOfFaunaActors: [Dinosaur]` //To know when events happen.

`ListOfFungiActors: [Fungi]` //To know when events happen.



Ecosystem Diagram